

Hybrid resolution refinement and the rodeo algorithm for eigenstate preparation: a demonstration on the pairing Hamiltonian

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The efficient preparation of correlated eigenstates is a central bottleneck for quantum simulation: variational quantum eigensolvers (VQE) suffer from exponentially vanishing gradients, adiabatic state preparation requires evolution times that diverge when the spectral gap closes, and projective filters such as quantum phase estimation and the rodeo algorithm converge exponentially only when the input state already has a sizeable overlap with the target eigenstate. We demonstrate, end to end and on a single controlled testbed — the constant-pairing Hamiltonian — a hybrid pipeline that resolves these tensions: a cheap low-resolution VQE state is prolonged into a larger model space, adiabatically *refined* across resolutions, and finally polished by the rodeo algorithm. Resolution refinement keeps the path gapped, so the adiabatic time scales only as $T \sim 1/\Delta E$ and the overlap p with the high-resolution ground state rises toward unity; the rodeo algorithm then removes the residual error exponentially, with a post-selected infidelity $1 - \mathcal{F}_M \lesssim \frac{1-p}{p} 2^{-M}$ in the number of cycles M and a post-selection acceptance that stays of order p . For four particles refined up to four levels ($k = 4$, eight qubits) the refined input ($p \simeq 0.998$) is filtered to a fidelity $1 - \mathcal{F} \sim 10^{-6}$ within one or two cycles, while an unrefined prolonged guess ($p \simeq 0.87$) still reaches $\sim 10^{-4}$ in a handful of cycles. All gate-level building blocks (UCCSD state preparation, the prolongation embedding, and the ancilla-based rodeo cycle) are implemented explicitly and validated against exact diagonalisation. A gate-count analysis quantifies the depth advantage of the hybrid: a single UCCSD-VQE preparation at $k = 4$ already costs 1312 CNOTs and must be re-executed throughout a classical optimisation loop, whereas one rodeo cycle costs a few hundred CNOTs per Trotter step, carries no variational parameters, and converges exponentially. The construction is the pairing-model realisation of the EIGEN-Q proposal and a concrete template for hardware demonstrations on mid-circuit-measurement platforms such as the Magne neutral-atom quantum computer.

I. INTRODUCTION

Determining the low-lying eigenvalues and eigenvectors of a many-body Hamiltonian is one of the grand challenges of quantum many-body theory, where the Hilbert-space dimension grows exponentially with the number of constituents. Quantum computers offer new routes to eigenvalue estimation and eigenstate preparation, but the state-of-the-art methods each face an obstacle at scale. The variational quantum eigensolver (VQE) [1] is limited by barren plateaus — gradients that vanish exponentially with system size [2]. Adiabatic state preparation requires evolution times that diverge if the gap between the tracked state and the rest of the spectrum closes along the path [3]. Projective spectral filters — quantum phase estimation [4] and the rodeo algorithm (RA) [5, 6] — converge exponentially in the number of cycles, but only once the initial state has a significant squared overlap p with the target eigenstate; their cost grows as $1/p$, and p itself typically falls exponentially

with system size.

These limitations are complementary, which suggests a hybrid strategy. *Resolution refinement* [7] bootstraps an eigenstate across model-space scales: an eigenstate of a coarse (low-resolution) Hamiltonian is lifted by a prolongation operator into a finer space and adiabatically evolved to the fine-resolution eigenstate. Because refinement preserves the low-energy spectral structure, the path stays gapped and the required adiabatic time scales only as the inverse physical gap (times a slowly growing system-size factor), delivering an output with $p \approx 1$. That output is exactly the high-overlap input the rodeo algorithm needs to converge exponentially. This is the core idea of the EIGEN-Q pipeline: (i) prepare a low-resolution eigenstate; (ii) prolong and adiabatically refine to a final time T^* ; (iii) apply the rodeo algorithm to reach the target fidelity and observables with exponential convergence.

In this work we realise the full pipeline end to end on a transparent and exactly solvable testbed, the constant-pairing Hamiltonian familiar from nuclear and condensed-matter many-body theory [17]. Working on a model whose exact spectrum is available at every stage lets us validate each component quantitatively: the gate-

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level UCCSD coarse solver against full configuration interaction (FCI) and coupled-cluster doubles (CCD); the prolongation and adiabatic refinement against the exact high-resolution eigenstate; and the rodeo filter against the analytic success-probability formulas of Ref. [6]. All circuits are implemented explicitly as rotation and CNOT gates (and an ancilla-based rodeo cycle), so the

demonstration doubles as a reference implementation for hardware.

II. THE PAIRING MODEL AND ITS QUBIT ENCODING

We study the constant-pairing Hamiltonian

$$H = \delta \sum_{p=0}^{k-1} p \sum_{\sigma=\uparrow\downarrow} n_{p\sigma} - \frac{g}{2} \sum_{p,q=0}^{k-1} a_{p\uparrow}^\dagger a_{p\downarrow}^\dagger a_{q\downarrow} a_{q\uparrow}, \quad (1)$$

with k doubly degenerate single-particle levels of spacing δ (set to $\delta = 1$) and a pairing strength g . The interaction scatters time-reversed pairs between levels; it conserves the seniority (the number of unpaired particles), so the ground state of N paired particles lives entirely in the seniority-zero subspace. We map level p and spin σ to qubit $2p + \sigma$ (so a k -level problem uses $2k$ qubits) and encode the fermionic operators through the Jordan–Wigner transformation. The Hartree–Fock (HF) reference fills the lowest $N/2$ levels, with $E_{\text{HF}}(N=4, g) = 2 - g$.

Throughout we use $N = 4$ particles as the running example. The fixed- N sector dimension is $\binom{2k}{N}$, growing from 1 at $k = 2$ (a single filled determinant) to 70 at $k = 4$ and 1820 at $k = 8$; the seniority-zero ground-state problem is correspondingly compact, which makes the model an ideal multi-resolution laboratory. “Basis refinement” here means *increasing k at fixed N* : the number of single-particle levels (and qubits) grows while the particle number is held fixed.

III. THE HYBRID PIPELINE

A. Coarse-state preparation: gate-level UCCSD–VQE

The coarse eigenstate is prepared with a unitary coupled-cluster ansatz restricted to singles and doubles (UCCSD), Trotterised into a product of excitation circuits acting on the HF reference,

$$|\Psi(\theta)\rangle = \prod_{\mu} e^{\theta_{\mu}(\hat{\tau}_{\mu} - \hat{\tau}_{\mu}^{\dagger})} |\text{HF}\rangle. \quad (2)$$

Through the Jordan–Wigner map each anti-Hermitian generator becomes a sum of commuting Pauli strings, so

every factor compiles into a basis change, a CNOT ladder, a single $R_z(\theta)$ rotation, and the uncompute; a single excitation yields two Pauli-rotation sub-circuits and a double excitation yields eight (the circuits are shown in Fig. 7). We optimise with the exact analytic gradient of the product ansatz (the naive two-term parameter shift is incorrect beyond the first factor, because later factors acquire frequency-one cross terms). Because the pairing force does not couple to seniority-changing single excitations, all singles amplitudes vanish to machine precision and the solver is effectively a *doubles* theory, mirroring the classical CCD structure.

B. The prolongation operator

The prolongation P embeds a low-resolution state into the high-resolution space. For basis refinement it maps every many-body basis state of the k_{low} space to the *same* occupation pattern in the k_{high} space, leaving the new single-particle levels empty. In canonical Jordan–Wigner ordering this is gate-free; for lattice refinement the analogous operator distributes each occupied coarse site over the neighbouring fine sites with two single-qubit rotations and two CNOTs per site per spatial direction, i.e. $O(N_{\text{sites}})$ gates per dimension [7]. The prolonged state $|\Phi_0\rangle = P|\psi_{\text{low}}\rangle$ satisfies $\langle\Phi_0|H_{\text{high}}|\Phi_0\rangle = E_{\text{low}}$ exactly, and already provides a high-quality first guess for the high-resolution ground state.

C. Resolution refinement (adiabatic phase)

The prolonged state is refined by adiabatically following the interpolated, energy-shifted Hamiltonian

$$H(s) = \cos^2\left(\frac{\pi s}{2}\right) P(H_{\text{low}} - \mu)P^\dagger + \sin^2\left(\frac{\pi s}{2}\right) (H_{\text{high}} - \mu), \quad s = \frac{t}{T}, \quad (3)$$

evolved by a Suzuki–Trotter product over total time

$T \leq T^*$. The shift μ (here $\mu = E_{\text{low}} + 0.6$) places the

tracked low-energy states below the large null space of $P^\dagger$, so the level followed from $s = 0$ to $s = 1$ is the high-resolution ground state. The instantaneous spectrum stays smooth and gapped along the entire path; consequently the adiabatic time scales as $T \sim 1/\Delta E$ rather than the $O(1/\Delta E_{\min}^2)$ of generic adiabatic preparation, and the overlap $p = |\langle \Psi_{\text{high}} | \Phi(T) \rangle|^2$ rises toward unity.

D. The rodeo algorithm

The rodeo algorithm [5, 6] removes the residual error left by the adiabatic tail. One cycle acts on the system register plus a single ancilla (reused across cycles by mid-circuit measurement; the circuit is shown in Fig. 8): a Hadamard on the ancilla, the controlled evolution e^{-iHt_k} of the system, a phase gate $P(Et_k)$ on the ancilla (which effectively shifts $H \rightarrow H - E$), a second Hadamard, and an ancilla measurement. Post-selecting the ancilla in $|0\rangle$ applies the non-unitary filter $\frac{1}{2}(\mathbb{I} + e^{iEt_k}e^{-iHt_k})$ to the system: an eigenstate $|E_j\rangle$ acquires the factor $e^{i(E-E_j)t_k/2} \cos[(E_j - E)t_k/2]$. For an eigenstate input the probability that all M ancilla measurements return 0 is therefore [6]

$$P_{0M}(E | \{t_k\}) = \prod_{k=1}^M \cos^2 \left[\frac{t_k}{2} (E_{\text{obj}} - E) \right], \quad (4)$$

and averaging the t_k over a Gaussian of RMS width σ gives

$$P_{0M}(E) = \left[\frac{1 + e^{-(E_{\text{obj}} - E)^2 \sigma^2 / 2}}{2} \right]^M. \quad (5)$$

On resonance ($E = E_{\text{obj}}$) the success probability is unity at every cycle, while off-resonant weight is suppressed geometrically toward the background 2^{-M} (the arithmetic mean of $\cos^2$ is $1/2$). Scanning E thus locates eigenvalues as peaks whose width $\sim 1/\sigma$ sets the energy resolution; fixing E at the target eigenvalue turns the algorithm into a state-preparation filter.

The controlled evolution is the only system-size-dependent ingredient. On hardware e^{-iHt} is realised by a Suzuki–Trotter decomposition over the $2k$ qubits [14–16]; for the single-qubit benchmark of Ref. [6] it reduces to a Z – Y – Z Euler rotation $R_{\hat{n}}(\theta) = e^{-i\theta \hat{n} \cdot \vec{\sigma}/2}$ implemented through the Qiskit native gate. In the simulations below we apply the controlled evolution exactly within the conserved N -particle sector, isolating the algorithmic behaviour from Trotter error (quantified in Sec. VI).

a. Convergence criterion. With E tuned to the target eigenvalue E_0 , the all-zero acceptance probability of an input $|\psi\rangle = \sum_k c_k |E_k\rangle$ after M cycles is

$$P_M = p + \sum_{k \neq 0} |c_k|^2 \left[\frac{1}{2} (1 + e^{-(E_k - E_0)^2 \sigma^2 / 2}) \right]^M, \quad p = |c_0|^2. \quad (6)$$

TABLE I. Ground-state energies at $g = 1$, $N = 4$, for the constant-pairing Hamiltonian. FCI is the exact N -sector result; CCD is coupled-cluster doubles; UCCSD is the gate-level circuit of Sec. III A. “#CNOT” is the two-qubit gate count of the UCCSD state preparation.

k	qubits	FCI	CCD	UCCSD	#CNOT
2	4	1.000000	1.000000	1.000000	0
3	6	0.794697	0.794697	0.794697	272
4	8	0.635548	0.630443	0.636987	1312
6	12	0.372664	0.350058	0.378719	–
8	16	0.145559	0.097263	0.157856	–

The target weight p is preserved at every round while off-resonant components decay geometrically, so $P_M \rightarrow p$ and the post-selected fidelity obeys

$$\mathcal{F}_M = \frac{p}{P_M} \simeq \left[1 + \frac{1-p}{p} 2^{-M} \right]^{-1}, \quad 1 - \mathcal{F}_M \lesssim \frac{1-p}{p} 2^{-M}. \quad (7)$$

Reaching fidelity $1 - \delta$ therefore needs only $M \gtrsim \log_2[(1-p)/(p\delta)]$ cycles, and since the expected number of repetitions to obtain an accepted run is $\sim 1/P_M \rightarrow 1/p$, the total controlled-evolution effort scales as $O(|\log \delta|/(p\epsilon))$ at energy resolution ϵ [5]. This is efficient precisely when resolution refinement supplies $p \approx 1$.

IV. RESULTS

A. Coarse solver: UCCSD versus exact references

Table I lists the ground-state energies at $g = 1$ for $N = 4$ as the basis grows from $k = 2$ to $k = 8$. The gate-level UCCSD circuit reproduces the operator-level (statevector) UCCSD to machine precision and stays variational ($E_{\text{UCCSD}} \geq E_{\text{FCI}}$), tracking FCI to $\sim 10^{-3}$ – 10^{-2} , whereas CCD over-binds. Figure 1 compares all solvers (exact, CCD, statevector and gate-level UCCSD, and a hardware-efficient ansatz [10]) as a function of k . The hardware-efficient ansatz is exact at $k = 2$ and reaches FCI at $k = 3$ only after a multi-start optimisation, and struggles at $k = 4$ — the symmetry-blind landscape that motivates the physically structured UCCSD circuit. Figure 2 shows that the same UCCSD solver remains a tight variational bound all the way to $k = 8$ (sixteen qubits), at the cost of a CNOT count that grows steeply (1312 CNOTs already at $k = 4$); the compiled circuits and their depths are analysed in Sec. V.

B. Resolution refinement: overlap and energy

At $N = 4$ the low-resolution ($k = 2$) space is a single filled determinant, so the prolonged state is the high-resolution HF reference, with first-guess overlaps 0.917

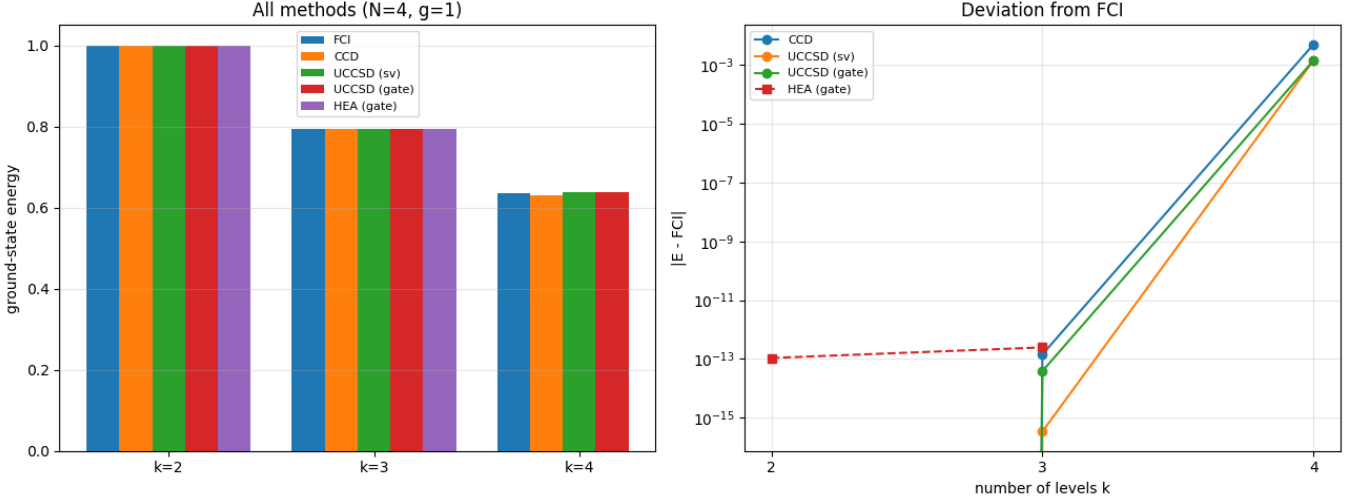


FIG. 1. Ground-state energy at $N = 4$, $g = 1$ obtained with all solvers (left) and their deviation from FCI (right), versus the number of levels k . The structured UCCSD circuits track FCI from above; CCD over-binds; the hardware-efficient ansatz matches FCI for $k \leq 3$ but loses accuracy at $k = 4$.

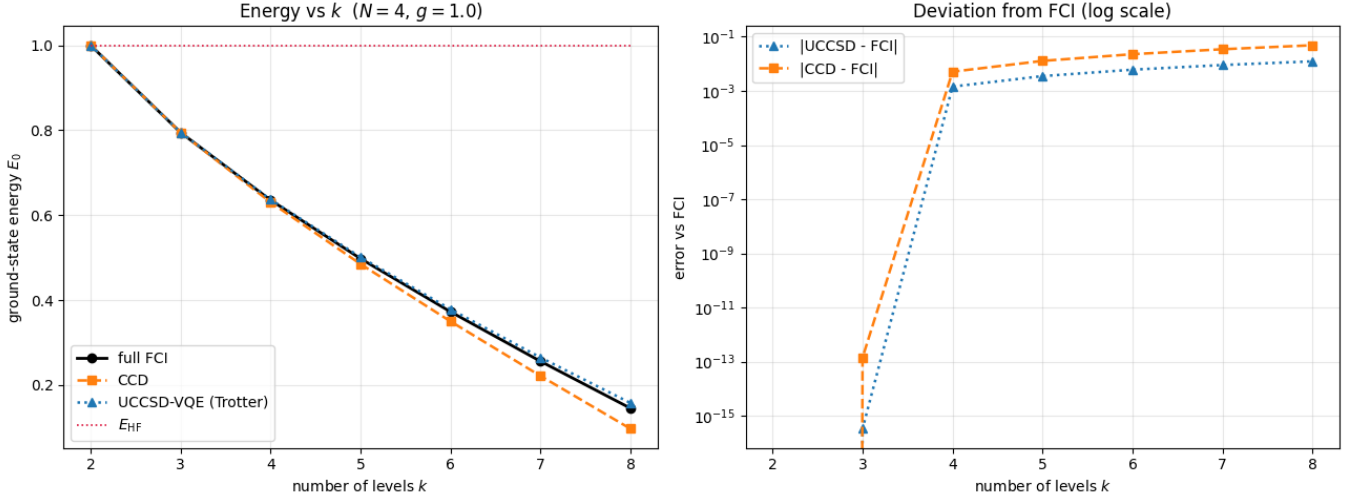


FIG. 2. Scalability of the Trotterised UCCSD-VQE: ground-state energy (left) and deviation from FCI (right) from $k = 2$ to $k = 8$ ($N = 4$, $g = 1$). The same solver remains variational and tight as the basis — and hence the qubit count — grows.

for refinement to $k = 3$ and 0.868 to $k = 4$. Adiabatic evolution along the gapped path of Eq. (3) drives both quantities toward their exact targets: the overlap rises to $\simeq 0.999$ and the energy falls from $E_{HF} = 1$ to the FCI value from above — 0.7964 for $k = 3$ (FCI 0.794697) and 0.6397 for $k = 4$ (FCI 0.635548) at $T = 30$ (Fig. 4). The instantaneous spectrum stays open along the whole path (Fig. 3, right), confirming the favourable $T \sim 1/\Delta E$ timescale; the larger $k = 4$ space starts lower in overlap and converges more slowly, exactly as expected from the system-size dependence of the refinement.

C. Rodeo filtering of the refined state

We verified the rodeo primitive in two ways: the explicit ancilla circuit agrees with the compact filter $\frac{1}{2}(\mathbb{I} + e^{iEt}e^{-iHt})$ to 10^{-16} , and the all-zero probabilities reproduce Eqs. (4)–(5), with the Gaussian-averaged single-cycle factor matching $\frac{1}{2}(1 + e^{-(\Delta E \sigma)^2/2})$ to three digits.

The state fed to the filter is the resolution-refined $k = 4$ state of Sec. IV B, with overlap $p = 0.998$ with the exact ground state; for contrast we also use the unrefined prolonged guess Φ_0 ($p = 0.868$). Scanning the target energy E (Fig. 5) produces a peak at the ground-state energy; a coarse scan ($\sigma = 2$) resolves it from the excited band,

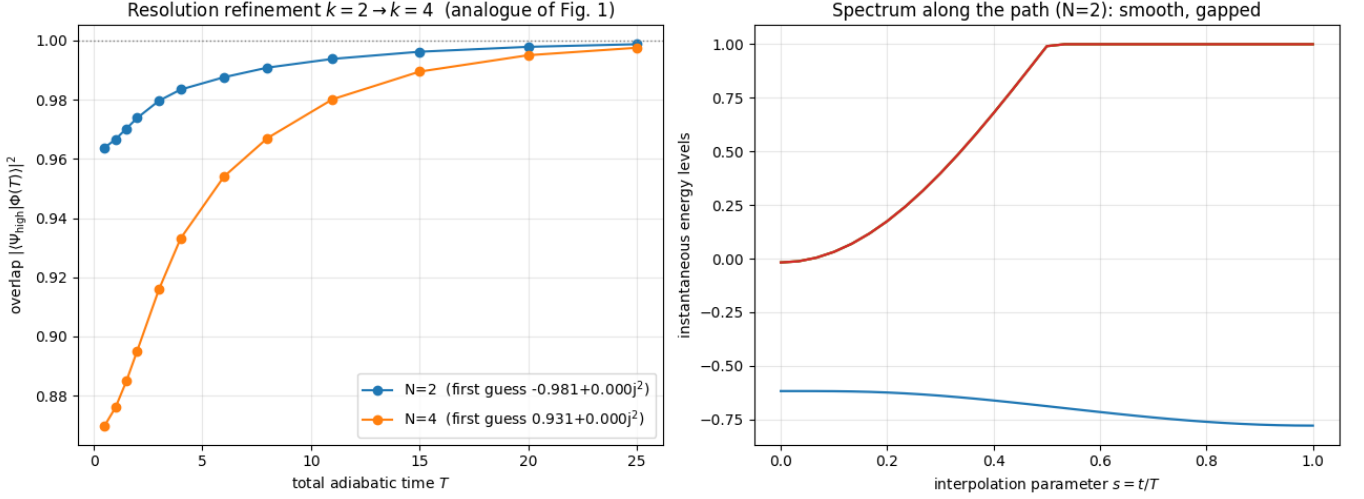


FIG. 3. Resolution refinement $k = 2 \rightarrow k = 4$. Left: overlap of the evolved state with the exact high-resolution ground state versus the total adiabatic time T , for $N = 2$ and $N = 4$. Right: the instantaneous spectrum along the interpolation parameter $s = t/T$ stays smooth and gapped, the condition for efficient refinement.

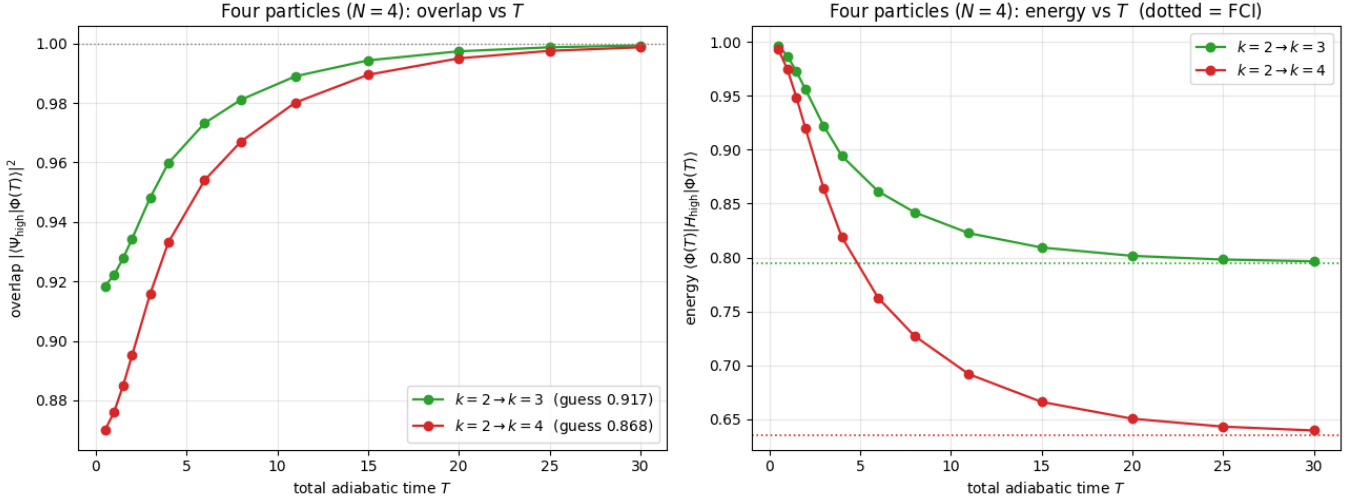


FIG. 4. Four-particle refinement up to $k = 4$: overlap (left) and physical energy $\langle \Phi(T) | H_{\text{high}} | \Phi(T) \rangle$ (right) versus T for $k_{\text{high}} = 3, 4$. The energy approaches the FCI value (dotted) from above while the overlap approaches unity, both on a $\sim 1/\Delta E$ timescale.

and a fine scan ($\sigma = 8$) locates it at $E = 0.6352$ against the exact $E_0 = 0.635548$. Starting from the refined state, essentially all spectral weight sits in the single ground-state peak; starting from Φ_0 , the small residual weight on excited states is visible as secondary structure above the 2^{-M} background.

Fixing $E = E_0$ and applying M cycles (Fig. 6), the post-selected fidelity rises exponentially and the cumulative acceptance settles at $\approx p$, in quantitative agreement with Eqs. (6)–(7). The refined input reaches an infidelity $1 - \mathcal{F} \simeq 5 \times 10^{-6}$ within one to two cycles; even the cruder prolonged input reaches $\simeq 1.3 \times 10^{-4}$ in a few cycles, while its acceptance holds at 0.868. The dotted reference lines $\frac{1-p}{p} 2^{-M}$ bracket the measured infidelities,

confirming the convergence criterion. This is the concrete pay-off of the pipeline: resolution refinement supplies the high overlap that turns the rodeo algorithm from a $1/p$ -expensive filter into a few-cycle polish.

V. QUANTUM CIRCUITS AND DEPTH COMPARISON

All quantum operations in this work are compiled to a common gate set of single-qubit rotations and CNOTs, so the resource costs of the two routes to the high-resolution ground state — a traditional full-space UCCSD-VQE versus the rodeo-terminated pipeline — can be compared

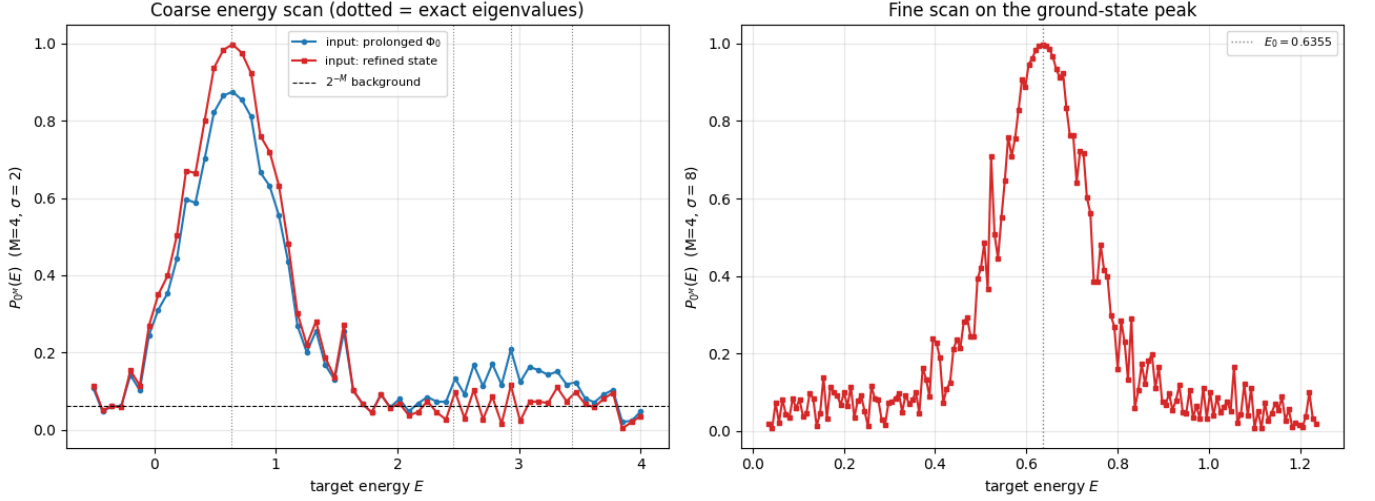


FIG. 5. Rodeo energy scan ($M = 4$). Left: coarse scan ($\sigma = 2$); the dotted vertical lines mark exact eigenvalues and the dashed line the 2^{-M} background. Right: fine scan ($\sigma = 8$) on the ground-state peak, located at $E = 0.6352$ (exact $E_0 = 0.6355$).

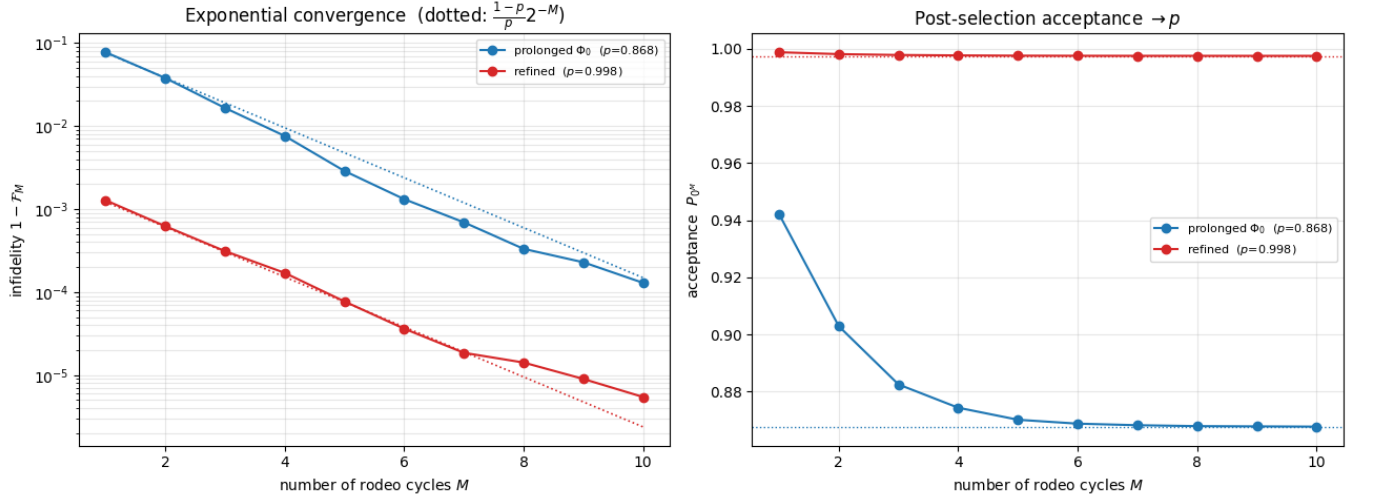


FIG. 6. Rodeo eigenstate preparation at $E = E_0$, averaged over random time sets. Left: post-selected infidelity $1 - \mathcal{F}_M$ versus the number of cycles M (dotted: the bound $\frac{1-p}{p}2^{-M}$). Right: the post-selection acceptance converges to the input overlap p .

gate by gate. Every number quoted below is produced (and verified against exact matrix exponentials) in the accompanying notebook.

A. Circuit building blocks

The elementary primitive is the Pauli rotation $e^{-i\frac{\phi}{2}P}$ for a Pauli string $P = \bigotimes_q \sigma_q$: each active qubit is rotated into the Z eigenbasis (H for X , $HS^\dagger$ for Y), a CNOT ladder accumulates the parity onto the last active qubit, a single $R_z(\phi)$ is applied, and the ladder and basis changes are uncomputed. A string of weight w therefore costs $2(w-1)$ CNOTs and one parametrised rotation; the circuit reproduces the exact exponential to 3×10^{-16} .

a. UCCSD preparation circuit. Through the Jordan–Wigner map a single excitation compiles into 2 Pauli rotations whose weight grows with the distance between the spin orbitals (8 CNOTs for adjacent levels, 16 for the $0 \rightarrow 4$ single at $k = 3$), and a generic double excitation into 8 rotations (48–64 CNOTs). The doubles that dominate the pairing problem are special: for a time-reversed pair moving between levels p and q the operators act on the adjacent qubits $(2p, 2p+1)$ and $(2q, 2q+1)$, so the internal Jordan–Wigner Z strings cancel and all 8 Pauli strings have weight 4 — 48 CNOTs per pair-hopping term *independent of the level separation*. The full state preparation at $N = 4$, $g = 1$ then costs 8 excitation operators, 40 rotations and 272 CNOTs at $k = 3$, and 26 operators, 160 rotations and

1312 CNOTs at $k = 4$ (Table I); at $k = 2$ the ansatz is empty (the sector is a single determinant) and the preparation is gate-free.

Figure 7 shows both levels of this hierarchy. Panel (a) is the VQE at the ansatz level: the product of excitation unitaries acts on the HF register, the energy is estimated from repeated measurements, and a classical optimiser updates the P amplitudes θ before the circuit is run again — it is this feedback loop, wrapped around the full-depth state preparation, that multiplies the 1312-CNOT cost into the total gate budget of the traditional route. Panel (b) is the compiled interior of a single weight-4 Pauli rotation, the sub-block that occurs eight times in every pair-hopping double and 160 times in the $k = 4$ preparation; its only parametrised element is the central $R_z(\phi)$, with ϕ fixed by the amplitude of the parent excitation operator.

b. Hardware-efficient ansatz. The HEA [10] used for contrast consists of L layers of R_y rotations on all $2k$ qubits interleaved with a CNOT entangler (nearest-neighbour pairs plus a wrap-around bond), i.e. $2k(L+1)$ variational angles and only $O(2kL)$ CNOTs: 12 angles at $k = 2$ ($L = 2$) and 30 at $k = 3$ ($L = 4$). Its depth is far smaller than UCCSD, but the circuit carries no particle-number or seniority structure; a number-penalty term and multi-start optimisation are needed at $k = 3$, and at $k = 4$ the optimiser fails to reach the ground state (Fig. 1). Depth is not the only currency: trainability of the symmetry-blind landscape becomes the bottleneck first.

c. Rodeo cycle. One rodeo cycle (Fig. 8) adds a Hadamard on the ancilla, the controlled evolution e^{-iHt_k} , the ancilla phase $P(Et_k)$ — the element that effectively shifts $H \rightarrow H - E$ — a second Hadamard, and a mid-circuit measurement; the single ancilla is reset and reused across the M cycles. The contrast with Fig. 7(a) is the structural point of this section: the rodeo circuit contains *no* variational parameters and closes no optimisation loop. The times t_k are drawn once from a Gaussian, the target energy E is a classical setting, and the algorithm is driven entirely by the measurement record — a run is accepted when all M ancilla outcomes read 0, which applies the filter of Sec. IIID cycle by cycle. Where the VQE re-executes its full-depth circuit to steer θ , the rodeo circuit is executed as printed, and its only system-size-dependent element is the controlled evolution. The controlled evolution is compiled from the same Pauli-rotation primitive: in each rotation sub-circuit only the central R_z needs to be controlled (if the rotation is absent the basis changes and CNOT ladder cancel), and a controlled R_z costs 2 extra CNOTs. For the pairing Hamiltonian (1) a first-order Trotter step contains the $k(k-1)/2$ pair-hopping terms (8 weight-4 rotations, 48 CNOTs each), k diagonal ZZ rotations (2 CNOTs each) and $2k$ single-qubit phases, i.e.

$$N_{\text{CNOT}}^{\text{step}}(k) = 24k(k-1) + 2k \quad (8)$$

TABLE II. Circuit-depth budget at $k = 4$ ($N = 4$, $g = 1$, eight system qubits). Top: the traditional route, a full-space gate-level UCCSD-VQE. Bottom: the stages of the hybrid pipeline (one repetition; the acceptance of the rodeo post-selection is $\approx p = 0.998$). n_s and n_T denote the number of Trotter steps in the refinement and in one controlled evolution, respectively; CNOT counts follow Table I and Eq. (8).

circuit	params	CNOTs	executions
UCCSD-VQE prep. ($k=4$)	26	1312	per eval. $\times$ shots
coarse UCCSD ($k=2$)	0	0	1
prolongation $k=2 \rightarrow 4$	0	0	—
refinement step ($k=4$)	0	296	n_s
rodeo cycle ($k=4$)	0	$\approx 420 n_T$	$M = 1-2$

CNOTs uncontrolled — 296 at $k = 4$ — and $\approx 4.2 \times 10^2$ CNOTs per step once the controls are added. A rodeo cycle with n_T Trotter steps therefore costs $\approx 4.2 \times 10^2 n_T$ CNOTs at $k = 4$.

B. Depth budget: UCCSD-VQE versus the rodeo-terminated pipeline

Table II summarises the comparison. The traditional route prepares the $k = 4$ ground state directly with the gate-level UCCSD-VQE: every energy or gradient evaluation inside the classical loop executes the full 1312-CNOT, 26-parameter circuit, each evaluation averaged over many shots, and the attainable accuracy saturates at the ansatz error, $E - E_{\text{FCI}} = 1.4 \times 10^{-3}$ at $k = 4$ (Table I). The cost also grows steeply with resolution — $272 \rightarrow 1312$ CNOTs from $k = 3$ to $k = 4$ — because the operator pool grows with both k and N .

The pipeline never runs a variational circuit at full resolution. Its only full-resolution circuits are (i) the parameter-free Trotter steps of the adiabatic refinement, 296 CNOTs each by Eq. (8), executed once per repetition with no optimiser attached, and (ii) $M = 1-2$ rodeo cycles of $\approx 4.2 \times 10^2 n_T$ CNOTs each. Even for several Trotter steps per cycle, a rodeo cycle remains a fraction of the depth of a single $k = 4$ UCCSD preparation — and it is executed a handful of times rather than throughout an optimisation loop, with the post-selection acceptance held at $p \simeq 0.998$ by the refinement. The pay-off in accuracy runs in the same direction: two rodeo cycles reach an infidelity of 5×10^{-6} (Fig. 6), three orders of magnitude below the variational floor of the deeper UCCSD circuit. The coherent depth of a single pipeline run is dominated by the refinement stage (n_s Trotter steps along the path); this cost is fixed and parameter-free, its length is set by the physical gap through $T \sim 1/\Delta E$, and the rodeo filter is self-correcting with respect to the residual Trotter error [6], so both n_s and n_T can be kept modest.

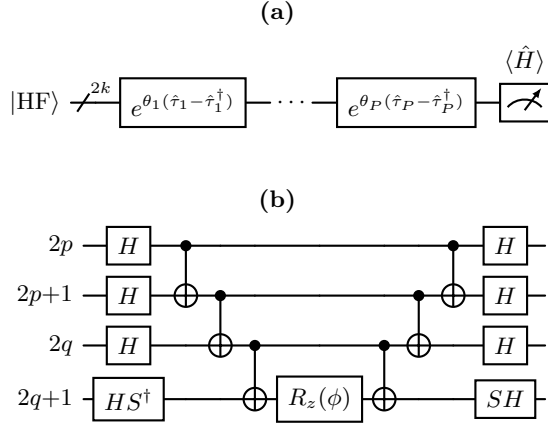


FIG. 7. The variational quantum eigensolver as a circuit. (a) Ansatz level: the Trotterised UCCSD product acts on the HF register ($2k$ qubits, slashed wire) and the energy is estimated by measuring $\langle \hat{H} \rangle$ (in practice a sum of Pauli expectation values, each averaged over many shots); a classical optimiser updates θ and the entire circuit is re-executed, once per energy or gradient evaluation. (b) Compiled circuit of one weight-4 Pauli rotation, here $e^{-i\frac{\phi}{2}X \otimes X \otimes X \otimes Y}$ acting on the four qubits of a pair-hopping double excitation $(2p, 2p+1) \rightarrow (2q, 2q+1)$: basis changes into the Z eigenbasis (H for X , $HS^\dagger$ for Y), a CNOT parity ladder, the single parametrised $R_z(\phi)$, and the uncompute. Eight such blocks (48 CNOTs) build one double excitation; 160 of them (1312 CNOTs) build the $k = 4$ preparation of Table I.

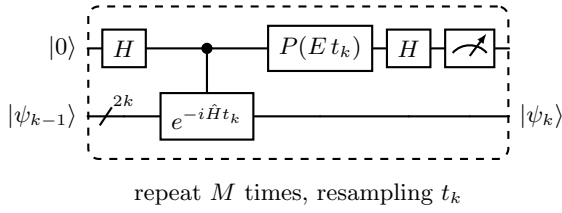


FIG. 8. One cycle of the rodeo algorithm: Hadamard on the ancilla, controlled time evolution $e^{-i\hat{H}t_k}$ of the $2k$ -qubit system register with a Gaussian-random time t_k , ancilla phase $P(Et_k)$, second Hadamard, and a mid-circuit measurement of the ancilla, which is then reset and reused for the next cycle. A run is accepted when all M outcomes read 0; each accepted cycle applies the filter $\frac{1}{2}(\mathbb{I} + e^{iEt_k}e^{-i\hat{H}t_k})$ to the system. Unlike the VQE of Fig. 7(a), the circuit carries no variational parameters and no classical optimisation loop; the controlled evolution, compiled from the same Pauli-rotation blocks as Fig. 7(b) with only the central R_z controlled, is the sole system-size-dependent element ($\approx 4.2 \times 10^2$ CNOTs per Trotter step at $k = 4$).

C. Remarks on adaptive and improved VQE

The steep UCCSD gate count is not irreducible, and much recent work aims precisely at shallower variational circuits (see Ref. [13] for a review). ADAPT-VQE [11] grows the ansatz one operator at a time, at each step appending the pool operator with the largest energy gradient and re-optimising all amplitudes; it typically reaches a target accuracy with a small fraction of the full UCCSD operator count and correspondingly shallower circuits. The pairing model makes the mechanism transparent: because the Hamiltonian conserves senior-

ity and the reference is seniority-zero, the gradients of all singles and of all pair-breaking doubles vanish identically, so an adaptive selection would retain only the $\frac{N}{2}(k - \frac{N}{2})$ pair-hopping doubles — four operators at $k = 4$, $N = 4$, i.e. $4 \times 48 = 192$ CNOTs in place of the 1312 of the full 26-operator ansatz, a sevenfold compression, with the discarded amplitudes expected to vanish at the optimum just as the singles do. Qubit-ADAPT-VQE [12] pushes further by drawing the pool from individual Pauli strings, removing the eightfold string overhead per fermionic double. These developments are complementary to the present pipeline rather than competing with it: the pipeline is agnostic to how the coarse state is produced, so an ADAPT-style coarse solver slots in directly, reducing the (already modest) low-resolution cost and potentially raising the prolonged overlap; conversely, no variational compression by itself supplies the exponential, optimiser-free convergence that the rodeo filter provides once the overlap is large.

VI. DISCUSSION AND OUTLOOK

The pairing model lets us exhibit the entire EIGEN-Q pipeline with every stage benchmarked against exact results: a gate-level UCCSD coarse solver, a gate-free prolongation, a gapped adiabatic refinement with $T \sim 1/\Delta E$, and a rodeo filter with $1 - \mathcal{F}_M \lesssim \frac{1-p}{p} 2^{-M}$. The two algorithms are genuinely complementary — refinement provides the large overlap $p \approx 1$ that the rodeo algorithm requires, and the rodeo algorithm removes the slow adiabatic tail that refinement alone leaves behind.

Moving to hardware changes only the controlled evolution. On a device e^{-iHt} is Trotterised over the $2k$ qubits,

which introduces an effective Hamiltonian whose eigenvalues differ from the exact ones at a controllable order in the step size; as noted in Ref. [6], the rodeo algorithm filters the eigenstates of this effective Hamiltonian, and its self-correcting design (small unitary errors broaden the energy window rather than biasing the estimate) makes it unusually robust to such errors. The mid-circuit measurement and deep coherent circuits required by the two phases are supported by neutral-atom architectures such as the Magne system, making the pipeline a natural target for a hardware demonstration.

Beyond the pairing model, the same construction applies to the targets of the EIGEN-Q proposal — light and medium-mass nuclei benchmarked against coupled-cluster and in-medium similarity renormalization group calculations, molecular systems toward chemical accuracy, and Hubbard models refined from a coarse to a fine lattice — with a closed-loop control layer that can adapt the refinement schedule from the rodeo acceptance statistics read out mid-circuit. The present results establish, on a fully controlled example, that the hybrid of resolu-

tion refinement and the rodeo algorithm prepares correlated eigenstates with exponential accuracy at a cost set by an overlap that the refinement itself makes large.

CODE AVAILABILITY

The complete implementation — pairing Hamiltonian, gate-level UCCSD–VQE, prolongation, adiabatic refinement, and the ancilla-based rodeo algorithm — is provided as a single Jupyter notebook that reproduces every figure in this article.

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- [1] A. Peruzzo, J. McClean, P. Shadbolt, M.-H. Yung, X.-Q. Zhou, P. J. Love, A. Aspuru-Guzik, J. L. O’Brien, *A variational eigenvalue solver on a photonic quantum processor*, Nat. Commun. **5**, 4213 (2014).
 - [2] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, H. Neven, *Barren plateaus in quantum neural network training landscapes*, Nat. Commun. **9**, 4812 (2018).
 - [3] T. Albash, D. A. Lidar, *Adiabatic quantum computation*, Rev. Mod. Phys. **90**, 015002 (2018).
 - [4] D. S. Abrams, S. Lloyd, *Quantum algorithm providing exponential speed increase for finding eigenvalues and eigenvectors*, Phys. Rev. Lett. **83**, 5162 (1999).
 - [5] K. Choi, D. Lee, J. Bonitati, Z. Qian, J. Watkins, *Rodeo algorithm for quantum computing*, Phys. Rev. Lett. **127**, 040505 (2021).
 - [6] Z. Qian, J. Watkins, G. Given, J. Bonitati, K. Choi, D. Lee, *Demonstration of the rodeo algorithm on a quantum computer*, Eur. Phys. J. A **60**, 151 (2024).
 - [7] S. Bogner *et al.*, *Quantum state preparation with resolution refinement*, Phys. Lett. B **875**, 140363 (2026).
 - [8] D. Lee, *Quantum techniques for eigenvalue problems*, Eur. Phys. J. A **59**, 275 (2023).
 - [9] J. Lee, W. J. Huggins, M. Head-Gordon, K. B. Whaley, *Generalized unitary coupled cluster wavefunctions for quantum computation*, J. Chem. Theory Comput. **15**, 311 (2019).
 - [10] A. Kandala, A. Mezzacapo, K. Temme, M. Takita, M. Brink, J. M. Chow, J. M. Gambetta, *Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets*, Nature **549**, 242 (2017).
 - [11] H. R. Grimsley, S. E. Economou, E. Barnes, N. J. Mayhall, *An adaptive variational algorithm for exact molecular simulations on a quantum computer*, Nat. Commun. **10**, 3007 (2019).
 - [12] H. L. Tang, V. O. Shkolnikov, G. S. Barron, H. R. Grimsley, N. J. Mayhall, E. Barnes, S. E. Economou, *Qubit-ADAPT-VQE: an adaptive algorithm for constructing hardware-efficient ansätze on a quantum processor*, PRX Quantum **2**, 020310 (2021).
 - [13] J. Tilly, H. Chen, S. Cao, D. Picozzi, K. Setia, Y. Li, E. Grant, L. Wossnig, I. Rungger, G. H. Booth, J. Tennyson, *The variational quantum eigensolver: a review of methods and best practices*, Phys. Rep. **986**, 1 (2022).
 - [14] H. F. Trotter, *On the product of semi-groups of operators*, Proc. Am. Math. Soc. **10**, 545 (1959).
 - [15] M. Suzuki, *Generalized Trotter’s formula and systematic approximants of exponential operators*, Commun. Math. Phys. **51**, 183 (1976).
 - [16] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, S. Zhu, *Theory of Trotter error with commutator scaling*, Phys. Rev. X **11**, 011020 (2021).
 - [17] M. Hjorth-Jensen, M. P. Lombardo, U. van Kolck (eds.), *An Advanced Course in Computational Nuclear Physics*, Lecture Notes in Physics **936**, Springer (2017).
 - [18] M. A. Nielsen, I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press (2010).